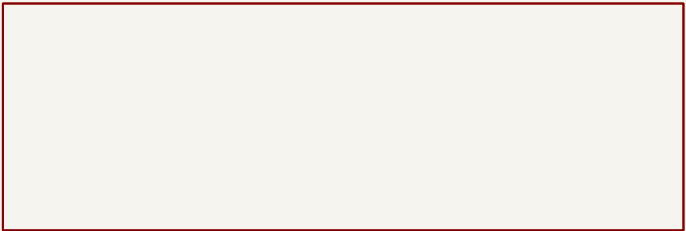


BATMAN / LiFePO4 8S + lead-acid 24 V

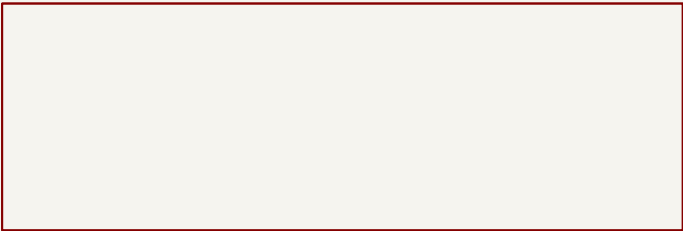
Four PCBs: A ESP + two I2C ports, B analog filters, C 3V3 distribution, D 5V distribution.
Two external ADS1115: analog from B, power/I2C from A. B,Cu only; WC = external cables.
LOAD bus C1/C2 stay externally wired. Prototype: verify sensor model, fit and current budgets.

power



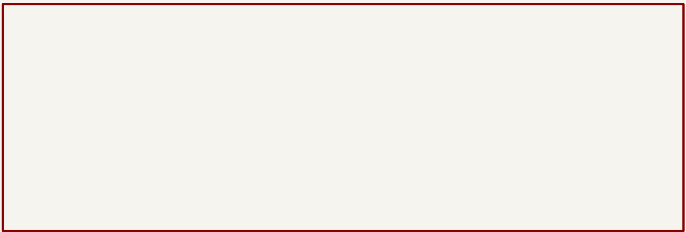
Файл: power.kicad_sch

charge



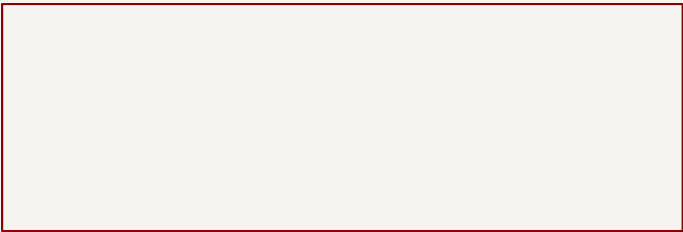
Файл: charge.kicad_sch

sense



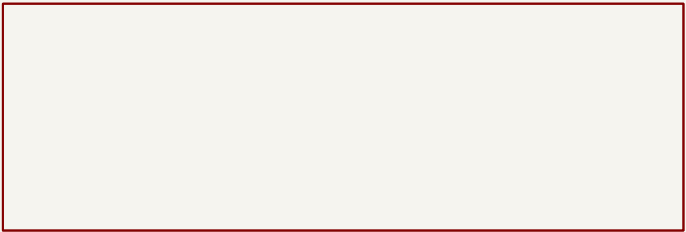
Файл: sense.kicad_sch

logic



Файл: logic.kicad_sch

supply3



Файл: supply3.kicad_sch

supply5



Файл: supply5.kicad_sch

Sheet: /		
File: batman.kicad_sch		
Title: BATMAN – dual battery manager		
Size: A4	Date: 2026–09–09	Rev: A–prototype
KiCad E.D.A. 10.0.6		Id: 1/7

BATMAN / POWER

P01–P28: external power wiring 2.5 mm2; P28 logic ground 0.75 mm2.
B1/B2 are AFTER sensors, BEFORE switches, IP– faces battery / source.
LOAD+/GND: external terminals, C1/C2 wired directly, no buffer PCB. D3 DNP; not a 35 V clamp.

A

B

C

D

E

F

A

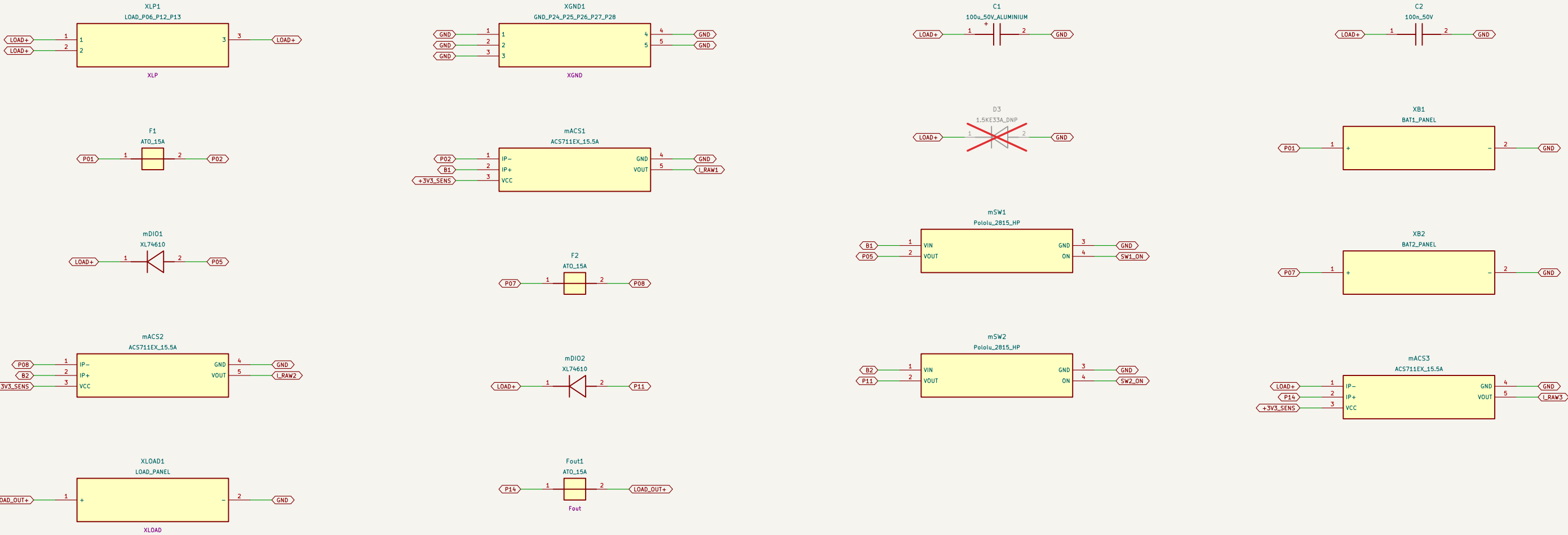
B

C

D

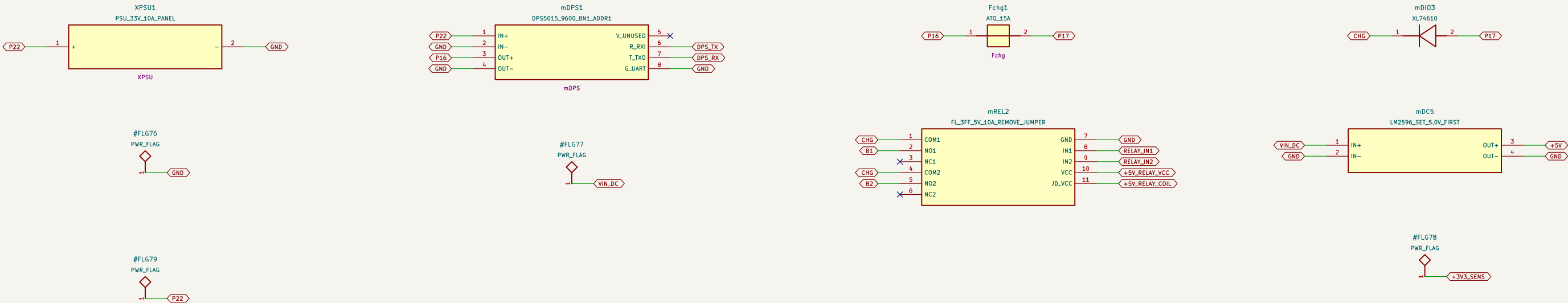
E

F



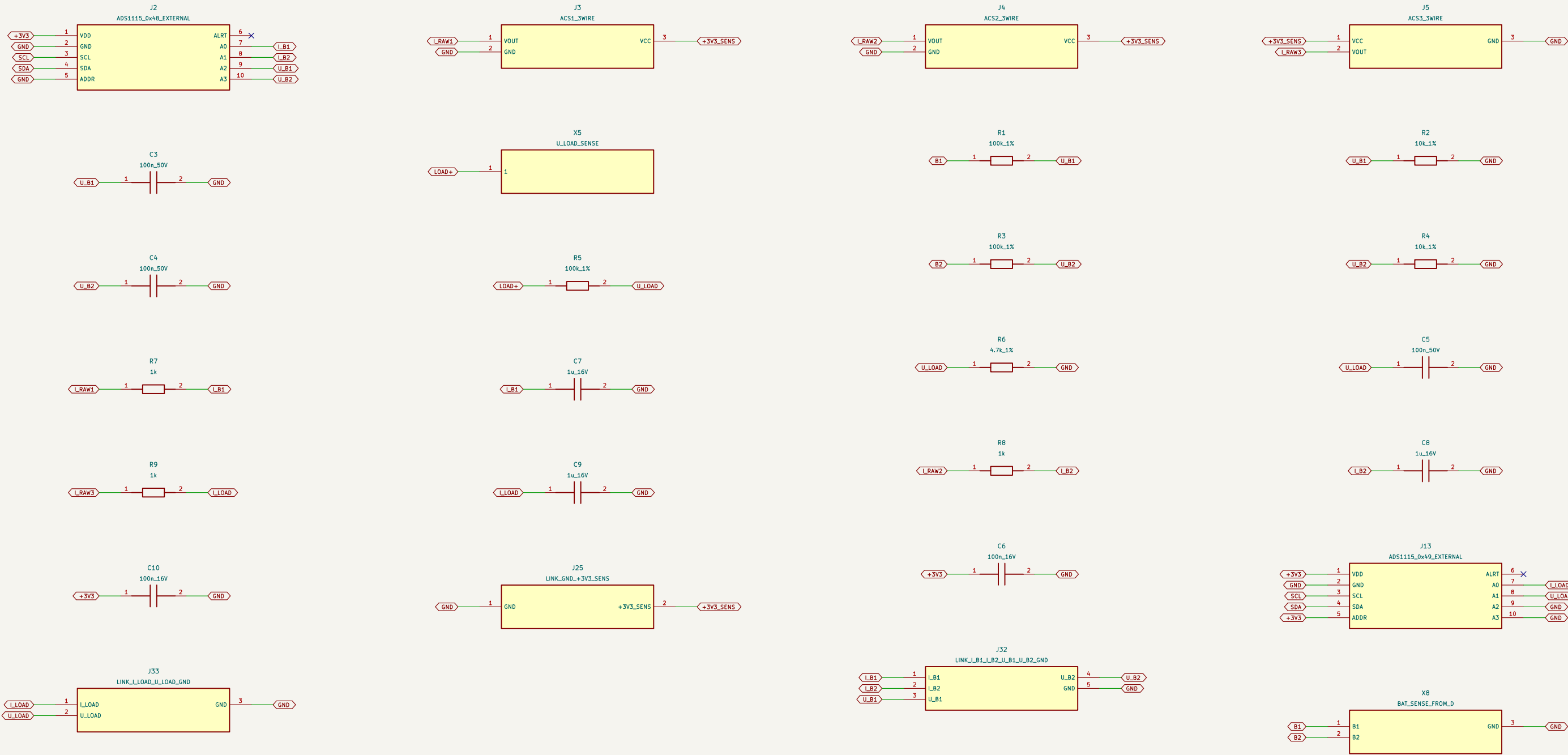
BATMAN / CHARGE

REMOVE relay VCC-JD_VCC jumper. Route VCC / JD_VCC separately from X3.
DPS: GND TX RX -> G R T; leave V unconnected. Set mDC5 to 5.0 V BEFORE plugging J1.
Never close K1 and K2 together. Switch relays only at verified zero current. Hardware rating: 10 A maximum.



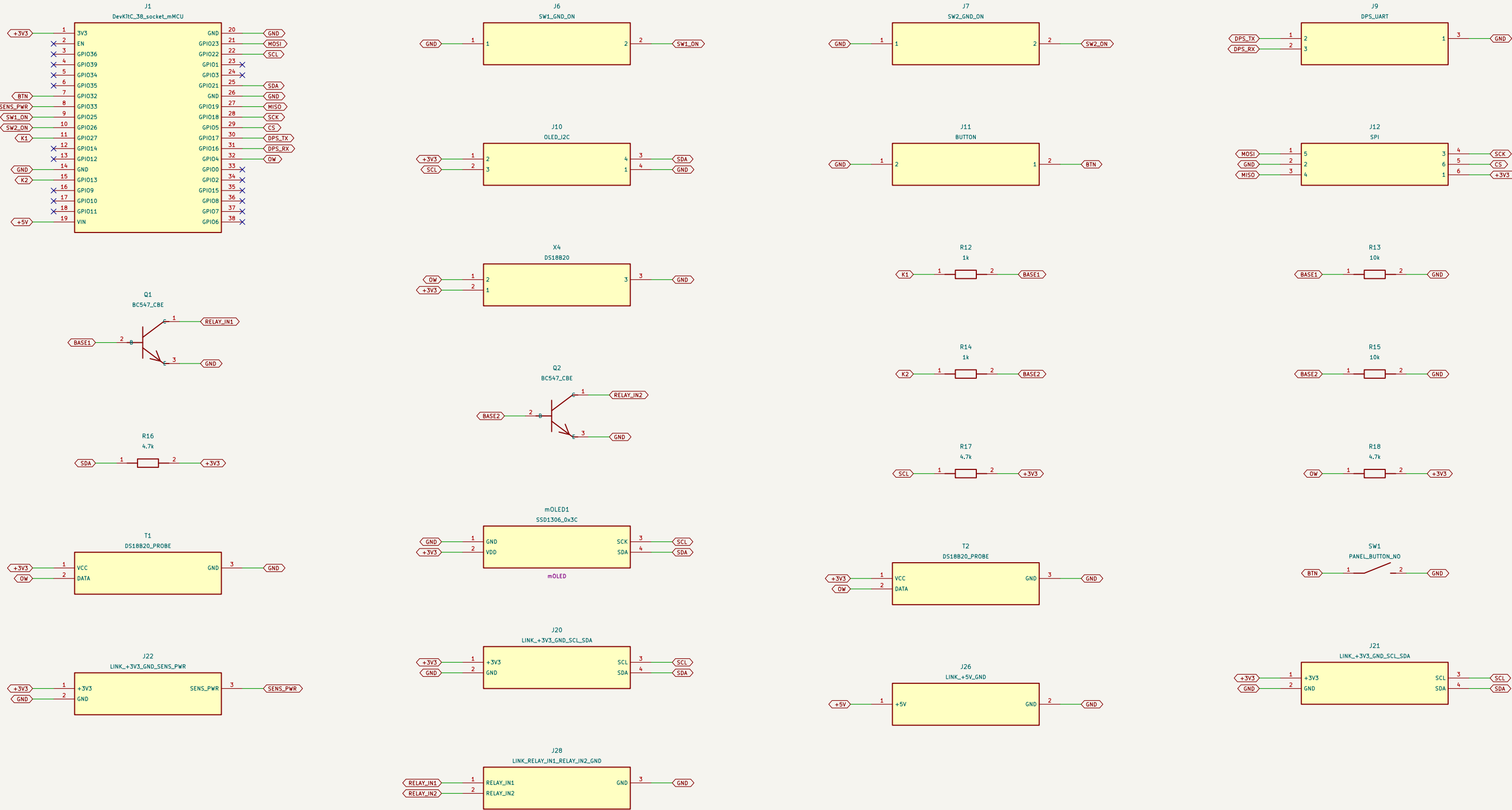
BATMAN / SENSE

PCB B; analog filters ONLY. J32 -> external J2 A0-A3/GND; J33 -> external J13 A0/A1/GND.
J2 ADDR=GND (0x4B); J13 ADDR=VDD (0x49). A2/A3=GND; both ALERT_NC. C6/C10 at modules.
ADCs powered from A, NOT B. J25: GND/3V3_SENS from C. No SDA/SCL on B. Verify sensor model.



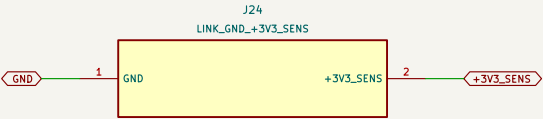
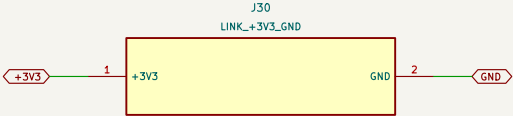
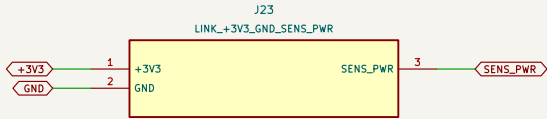
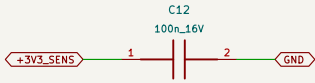
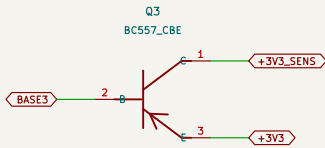
BATMAN / LOGIC

PCB A: J20 -> external ADS J2; J21 -> external ADS J13. Both: 3V3/GND/SCL/SDA.
Shared GPIO21=SDA/GPIO22=SCL; one pull-up pair: R16/R17; GPIO34/39 NC.
J22 -> C, J26 receives 5 V from D; J28 relay control to D. Cable map: output/cable-map.csv.



BATMAN / SUPPLY3

PCB C: DISTRIBUTION ONLY, no regulator. J23 receives 3V3/GND/SENS_PWR from ESP PCB A.
J24 supplies PCB B sensors: GND/3V3_SENS. J30 is a spare 3V3/GND output.
Q3 switches sensor supply only. Do NOT connect another regulator to this 3V3 rail.



BATMAN / SUPPLY5

PCB D: DISTRIBUTION ONLY. X3 receives regulated 5.0 V from external LM2596.
X1 battery taps feed diode OR -> X2 LM2596 Input; X6 -> X8 on ADC. PCB B (sense only).
NT1 star: separate relay VCC/JD-VCC. REMOVE relay jumper. J27 -> ESP; J31 spare 5 V.

A

B

C

D

E

F

A

B

C

D

E

F

